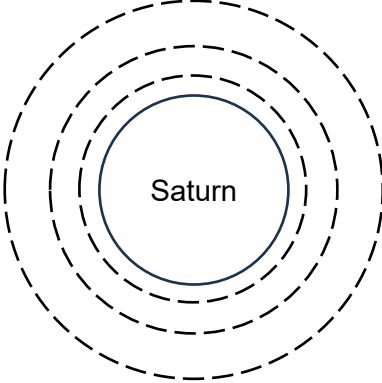


Qn	Suggested Solutions
8(a)	A <i>geostationary orbit</i> is one where the satellite will remain in the same position in the sky relative to the Earth's surface.
(b)(i)	$\text{angular velocity} = \frac{v}{r}$ $= \frac{2.26 \times 10^4 (2)}{1.49 \times 10^8}$ $= 3.03356 \times 10^{-4}$ $= 3.03 \times 10^{-4} \text{ rad s}^{-1} \text{ (3sf)}$
(ii)	$T = \frac{2\pi}{\omega}$ $= \frac{2\pi}{3.03356 \times 10^{-4}}$ $= 20712 \text{ s}$ $\approx 5 \text{ h } 45 \text{ min}$
	Since orbital period of particle is not equal to the rotational period of Saturn (10 h and 14 min), the particle is not in a stationary orbit.
(iii)	Gravitational force of Saturn provides centripetal force.
	$\frac{GMm}{r^2} = mr\omega^2$
	$r^3\omega^2 = GM$
(iv)	$M = \frac{r^3\omega^2}{G}$ $= \frac{(0.745 \times 10^8)^3 (3.03356 \times 10^{-4})^2}{6.67 \times 10^{-11}}$
	$= 5.70490 \times 10^{26}$
	$= 5.7 \times 10^{26} \text{ kg (2sf) (shown)}$
(c)(i)	<i>Gravitational potential</i> at a point in a gravitational field is defined as the work done per unit mass by an external agent
	in bringing a small test mass from infinity to that point, without producing any acceleration.

(ii)	
(iii)	To escape Saturn's gravitational field, the particle must gain GPE and reach infinity with $KE \geq 0$. OR Applying conservation of energy, Gain in GPE = Loss in KE $GPE_{\infty} - GPE_i = KE_i - KE_{\infty}$
	$GMm\left(\frac{1}{r_i} - \frac{1}{r_{\infty}}\right) = \frac{1}{2}m(v_i^2 - v_{\infty}^2)$ $v_i = \sqrt{(2 \times 6.67 \times 10^{-11} \times 5.7 \times 10^{26})\left(\frac{1}{0.745 \times 10^8} - 0\right) + 0}$
	$v_i = 3.19475 \times 10^4$ additional v required $= 3.19475 \times 10^4 - 2.26 \times 10^4$ $= 9347.5$ $= 9350 \text{ m s}^{-1} \text{ (3sf)}$
(d)(i)	Let the distance be x. $\frac{GM_T}{x^2} = \frac{GM_S}{(1.22 \times 10^9 - x)^2}$
	$\frac{1.4 \times 10^{23}}{x^2} = \frac{5.7 \times 10^{26}}{(1.22 \times 10^9 - x)^2}$ $(1.22 \times 10^9 - x)\sqrt{1.4 \times 10^{23}} = 5.7 \times 10^{26}x$
	$x = 1.88249 \times 10^7$ $= 1.88 \times 10^7 \text{ m (3sf)}$

